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9.3.3 The Inverse Power Method

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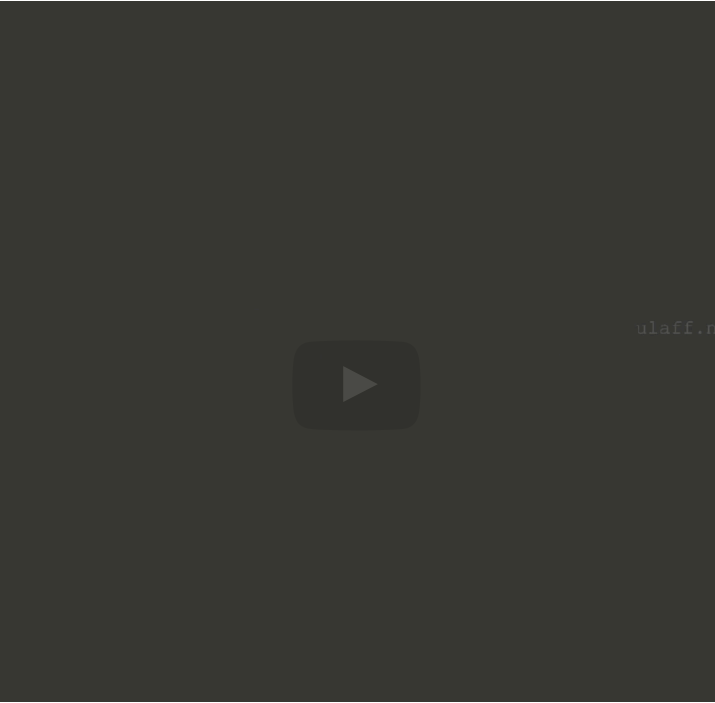
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eigenvector. And to come up with a method for that, we are going to exploit the relationship between the eigenvalues and eigenvectors of the matrix and the eigenvalues and eigenvectors of the inverse of that matrix, if the matrix of course is nonsingular. So let's assume that a is nonsingular. In a homework we're going to ask you the question: If we know an eigenvalue and eigenvector of A, how can we

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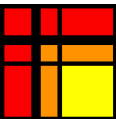
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Reading assignment

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Read Unit 9.3.3 of the notes. [\[LINK\]](#)

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Homework 9.3.3.1

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Let $A \in \mathbb{C}^{m \times m}$ be nonsingular, and (λ, x) an eigenpair of A .

Which of the follow is an eigenpair of A^{-1}

- ☐ (λ, x)
- ☐ $(\lambda, A^{-1}x)$
- ☐ $(1/\lambda, A^{-1}x)$
- ☒ $(1/\lambda, x)$



Solution

For a complete solution, see the [notes](#).

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Answers are displayed within the problem

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inverse. The lambda that shows up over here is just the inverse of the lambda that showed up when we diagonalized A. So what do we notice? This is just equal to the matrix that are the eigenvectors of matrix A. And then the diagonal matrix that consists of the values one over lambda_0 through one over lambda_{m-1}, and then the matrix X inverse.

Okay? And from that we can totally read off what we need. If we have m linearly

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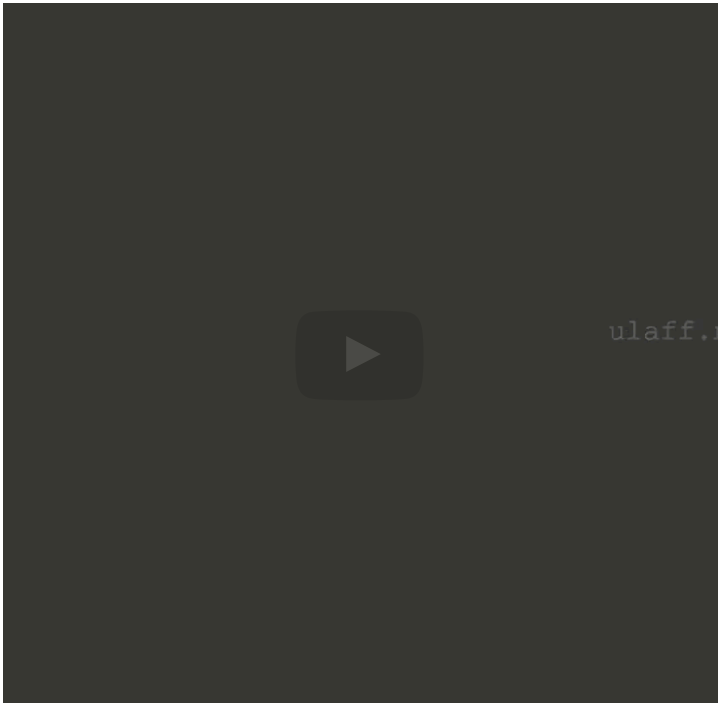
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dominate. Okay, and then we can express the exact same thing in matrix form as follows. In order to analyze this then what we would do is say, "Instead of dividing by λ_0 , let's multiply by λ_{m-1} ." In that case, we get a λ_{m-1} to the k th power here. This first term here is replaced by 1. And these terms are all replaced by the ratios between the smallest eigenvalue in magnitude and the next smallest eigenvalue the

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